
Current position

2025 – present **Associate Professor**, École Centrale de Nantes.

Education

- 2020 – 2024 **PhD in Applied Mathematics**, Université d'Évry Val d'Essonne & Politecnico di Torino, Évry & Turin.
Advisors: Nicolas Meunier and Luigi Preziosi.
Study of the influence of the cell nucleus on cell motility.
Joint international thesis.
Thesis defended on 20 September 2024.
- 2019 – 2020 **Master 2, Mathématiques pour les Sciences du Vivant**, Université Paris-Saclay, Orsay.
- 2018 – 2019 **Master 1 Mathématiques fondamentales**, Université Paris-Sud, Orsay.
- 2017 – 2020 **Magistère de mathématiques**, Université Paris-Saclay, Orsay.
- 2015 – 2018 **Licence de mathématiques**, Université Paris-Sud, Orsay.

Experiences

- 2024 – 2025 **Postdoctoral Researcher**, INRIA, Rennes.
Numerical methods for evolution in spatially structured population models.
Work in collaboration with Hélène Hivert.
Teaching (54 hours): Tutorial of general analysis and algebra for first year students in physics-chemistry and computer sciences.
- 2023–2024 **ATER**, Université d'Évry Paris Saclay, Évry
Teaching (80 hours): Tutorial of general analysis and algebra for first year students in physics-chemistry, biology and mathematics.
- 18 July - 26 August 2022 **CEMRACS 2022**, (*Centre d'été de mathématiques et de recherche avancée en calcul scientifique*), CIRM Marseille, Transport in Physics, Biology and Urban traffic.

Publication

- 2026 Claire Alamichel and Nicolas Meunier, *Existence of traveling wave for a coupled incompressible Darcy's free boundary model with undercooling effect and surface tension*, ESAIM: M2AN, 60 3 (2026) 1503-1524.
- 2024 Claire Alamichel, Juan Calvo, Erwan Hingant, Saoussen Latrach, Nathan Quiblier and Romain Yvinec, *Modeling compartmentalization within intracellular signaling pathway*, ESAIM: PROCEEDINGS AND SURVEYS, 77(2024) 100-122.

Given talks, and seminars

- 1 – 5 June 2026 **CANUM Congress**, *Saint-Jacut-de-la-Mer*, Modèles de dynamique évolutive de populations - modélisation, analyse et simulations numériques à l'aide d'un schéma AP, talk in a minisymposium.
- 18 – 22 May 2026 **Mathematical models for proliferation and propagation**, *CIRM, Marseille*, Study of the influence of the cell nucleus on cell motility: modelling and numerical simulations.
- 24 – 27 November 2025 **ReaDiNet2025 : A ReaDiNet workshop on deterministic and stochastic PDEs**, *Obernai*, Study of the influence of the cell nucleus on cell motility: modelling and numerical simulations.
- 25 September 2025 **GT BioMaths Normand**, *Rouen*, Study of the influence of the cell nucleus on cell motility: modelling and numerical simulations.
- 2 – 6 June 2025 **SMAI Congress**, *Carcans Maubuisson*, Rôle du noyau sur la motilité cellulaire : modélisation, étude mathématique et simulations numériques.
- 25 February 2025 **Séminaire d'Analyse, Phénomènes Stochastiques et Applications**, *Brest*, Influence du noyau sur la motilité cellulaire : modélisation et simulations numériques.
- 6 February 2025 **Séminaire de Calcul Scientifique et Modélisation**, *Bordeaux*, Influence du noyau sur la motilité cellulaire : modélisation et simulations numériques.
- 30 January 2025 **Séminaire d'Analyse**, *Tours*, Influence du noyau sur la motilité cellulaire : modélisation et simulations numériques.
- 14 January 2025 **Séminaire de Mathématiques Appliquées**, *Nantes*, Étude de l'influence du noyau sur la motilité cellulaire.
- 16 July 2024 **Journal Club DISMA**, *Turin*, Influence of the nucleus on cell motility: modelling and numerical simulations.
- 18 June 2024 **LaMME doctoral students' seminar**, *Évry*, Modélisation de la motilité cellulaire et simulations numériques.
- 13 June 2024 **Numerical analysis seminar**, *Rennes*, Modélisation de la motilité cellulaire et simulations numériques.
- 6 – 10 November 2023 **XIV Jornada de matemáticos ecuatorianos en Paris**, *Paris*, Modelling and numerical simulations of cell motility.
- 12 October 2023 **Presentation day for the LaMME to students in the Master MEEF programme at the Université Paris-Saclay**, *Évry*, Simulations numériques du problème de la migration cellulaire sur un disque avec une vitesse d'advection homogène dans l'espace, *mediation talk*.
- 9 May 2023 **IRMA doctoral students' seminar**, *Strasbourg*, A propos de la modélisation de la motilité cellulaire.
- 7 April 2023 **Orsay analysis doctoral students' seminar**, *Orsay*, Un modèle de la motilité cellulaire en présence d'un signal extérieur.

Attended conferences and workshops

- 15 – 17 June 2026 **Data assimilation and inverse problems for biomedical problems, Nantes.**
- 23 – 24 June 2025 **Journées Parité de la communauté mathématique 2025, Rennes.**
- 26 – 28 June 2024 **Journées Maths Bio Santé, Nantes.**
- 6 – 9 September 2022 **BioTOMath - Mathematical Challenges in Biology and Medicine, Turin.**
- 28 March – 1er April 2022 **Mathematical Modeling of Organization in Living Matter, CIRM Marseille.**
- 10 – 14 January 2022 **Workshop 'Tissue growth and movement', IHP Paris.**

Administrative tasks

- 2025 – present Member of the LMJL bureau.
- 2025 – present Co-organiser of the LMJL applied mathematics seminar.
- 2023 – 2024 Organiser of the LaMME doctoral students' seminar.
- 2022 – 2023 Elected doctoral student representative on the EDMH doctoral school board.

Skills

- Programming language FreeFem++, Python, C++, Julia.
- Languages French (native), English (advanced), German (scolar), Italian (notions).